PRODUCT REFERENCE MANUAL

UNIT ELECTRONICS · DEVLAB ECOSYSTEM · ATOM FAMILY

UNIT ATOM GT36537

MPN: UE0164

Version 1.0.0

UNIT ATOM GT36537 ambient light sensor top view



GT36537 ambient-light module with an onboard I2C interface, Qwiic expansion, and direct analog access.

GT36537 | I2C + ANALOG | QWIIC | DEVLAB ECOSYSTEM | ATOM FAMILY

DOCUMENT MAP

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Description

The DevLab I2C GT36537 LDR Ambient Light Sensor is an ambient-light sensor module developed by UNIT Electronics as part of the DevLab ecosystem. It combines a GT36537 photoresistor (LDR), wired as a voltage divider, with an onboard microcontroller-based interface controller, providing both I2C communication and direct access to the sensor's raw analog signal.

The board provides three I2C connectors sharing the same bus, a dedicated RAW Signal Header for direct access to the unprocessed sensor output, and power/status indicators. The onboard microcontroller reads the GT36537 divider signal through its analog-to-digital converter (ADC) and makes the resulting measurements available through the I2C interface.

Applications

- I2C-connected ambient-light acquisition
- Direct analog light measurement during development, testing, and characterization
- Automatic display and indicator brightness control
- Smart lighting and home/industrial automation
- Day/night and relative-light detection
- Environmental and IoT light-level data logging
- Educational I2C and photoresistor experiments
- Integration into I2C sensor networks

DevLab Format Compatibility

As part of the DevLab ecosystem, the GT36537 module follows a compact form factor intended for rapid prototyping and integration with other DevLab modules. The board provides three I2C connectors carrying `GND`, `VCC`, `SDA`, and `SCL`, all wired to the same bus, giving multiple physical access points to the I2C interface and supporting different module integration and mounting configurations.

Connector pitch, mechanical drawings, and supply-voltage limits for this GT36537 revision are still pending publication; verify the current hardware documentation before connecting the module to a host.

Hardware Features

- GT36537 ambient-light photoresistor, wired as a voltage divider
- Onboard microcontroller handling I2C communication and ADC acquisition
- 7-bit I2C slave operation using the DevLab Device Protocol (DDP)
- Three I2C connection positions carrying `GND`, `VCC`, `SDA`, and `SCL`, all connected to the same bus
- Dedicated RAW Signal Header exposing the unprocessed GT36537 divider output, independent of the I2C interface
- Power and user/status indicators
- Dedicated test points for debug

The controller implements DevLab Device Protocol (DDP) v1.0, using DevLab Interface for I2C bus management. DevLab Interface is integrated into the module's own library, so no separate bus-handling code is required. The GT36537 profile is identified by Device ID `0x0106`, and the factory I2C address is `0x26`.

Detailed electrical ratings, mechanical dimensions, and pinout diagrams for this GT36537 revision are still pending and will be published once the corresponding hardware assets are available.

1 The Board

The design supports two host paths. An I2C-capable controller can attach through any of the three I2C connectors, while a host or test instrument with an analog input can sample the dedicated RAW Signal Header directly. The GT36537 photoresistor (LDR) is wired as a voltage divider; its midpoint is read by the onboard microcontroller and made available through the I2C interface, and remains separately available as an unprocessed analog signal on the RAW Signal Header.

Detailed mechanical, pinout, and schematic assets for this GT36537 revision are still pending publication. The values in this chapter reflect the current confirmed design; items not yet released are marked accordingly.

1.1 Accessories

The module is supplied with a Qwiic-compatible cable for connection to the I2C interface.

Accessory	Purpose	Selection notes
JST/Qwiic-compatible cable	Connects GND, VCC, SDA, and SCL	Connector pitch and orientation for this GT36537 revision are pending publication; verify against the populated connector before use

1.2 Recommended Test Equipment

The following equipment may be used for integration, testing, and validation. These items are not included with the module.

Equipment	Purpose	Selection notes
I2C-capable host	Scans and communicates with the module	Use 7-bit addressing over the DevLab Device Protocol (DDP)
Analog test lead or carrier	Provides access to VCC, GND, and SIGNAL on the RAW Signal Header	SIGNAL must connect to a voltage-compatible ADC input
Logic analyzer	Checks I2C communication activity	Use input thresholds compatible with the powered board
Reference lux meter	Supports optical calibration and validation	Required for quantitative illuminance validation

1.3 Board Identification

Item	Value
Product	DevLab I2C GT36537 LDR Ambient Light Sensor
Brand / company	UNIT Electronics
Board ecosystem	DevLab
Product type	I2C-compatible ambient-light module with direct raw-signal access
Optical component	GT36537 photoresistor (LDR), wired as a voltage divider

Item	Value
Interface controller	32-bit Arm Cortex-M0+ microcontroller; exact fitted part pending confirmation
Manufacturer Part Number (MPN)	Pending
Current board artwork	Pending
Product Reference	Pending — this chapter reflects the GT36537 revision in progress

Board, pinout, schematic, and documentation revisions are controlled independently.

1.4 Board Components and Features

Component / Feature	Function	Implementation / Notes
GT36537 photoresistor (LDR)	Changes resistance with incident visible light	Wired as a voltage divider; primary ambient-light sensing element
Onboard microcontroller	Reads the divider signal through its ADC and manages I2C communication	32-bit Arm Cortex-M0+; exact fitted part pending confirmation
I2C connectors	Provide module power and I2C bus access	Three connectors, all wired to the same bus
RAW Signal Header	Provides direct access to `VCC`, `GND`, and the unprocessed sensor `SIGNAL`	Independent of the I2C interface; intended for analog measurement, testing, and characterization
Power indicator	Indicates that the board is powered	Onboard power LED
User / status indicator	Provides firmware-controlled status indication	Onboard status LED
Test points	Support debug and validation	Dedicated test points; assignment pending detailed publication

1.5 Board Layout and Reference Designators

Board-layout artwork, component placement, and reference designators for this GT36537 revision are still pending publication. The topology, top-view, and bottom-view images previously published for this repository belong to the prior TEMT6000 (UE0098) hardware revision and must not be used as a layout reference for the GT36537 module. This section will be completed once the corresponding hardware assets are released.

Refer to Chapter 4 for user connections once the released schematic and pinout are available.

1.6 Board Views

Top- and bottom-view images for this GT36537 revision are pending publication. A product photo and board-view images will be added once the corresponding hardware assets are available.

1.7 Handling

Use normal ESD precautions. Keep the sensor surface clean and optically unobstructed. Remove power before changing connectors.

2 Ratings

This chapter summarizes the electrical operating conditions and interface characteristics relevant to integrating the DevLab I2C GT36537 LDR Ambient Light Sensor. It covers the module supply, I2C operation, analog sensor path, and controller capabilities. Use these ratings to select a compatible power source and host interface and to identify the electrical characteristics that require application-specific verification.

2.1 Recommended Operating Conditions

Symbol	Description	Min.	Typ.	Max.	Unit
VCC	Module supply voltage	—	3.3 or 5	—	V
VI2C	I2C logic-high and pull-up voltage	—	VCC	—	V
fSCL	I2C clock frequency	100	—	400	kHz
VSIG	Direct analog SIGNAL voltage range	—	Not specified	—	V
ICC	Module supply current	—	Not specified	—	mA
TA	Module ambient operating temperature	Not specified	—	Not specified	°C

The VCC values are the supported nominal operating points; they are not complete-module absolute maximum ratings. The validated continuous supply range around each nominal point is not specified.

Use a current-limited source during initial bring-up. The I2C pull-ups follow the module supply, so the host must tolerate the actual bus voltage. When the module operates at 5 V with a 3.3 V host, bidirectional level translation may be required.

The direct SIGNAL range requires analog-stage verification and measurement. Module current consumption must include the controller, indicators, pull-ups, and sensor load. The complete-module ambient-temperature range has not been qualified. I2C pull-up resistance, bus capacitance, and complete-module absolute maximum ratings are also not specified.

2.2 Digital Interface and Firmware Characteristics

Parameter	Value	Description
I2C addressing	7-bit slave	Valid configurable addresses are 0x08..0x77
Factory I2C address	0x26 (7-bit)	Pass 0x26 directly to the host I2C library; 0x4C is the 8-bit write address, not the device address
Device protocol	DDP v1.0	Command transaction followed by an exact-length read transaction

Parameter	Value	Description
Logical Device ID	0x0106	GT36537 identity; independent of the I2C address
Firmware / hardware	1.0 / 1.0	Current observable controller profile
Capability bitmap	0x000001B9	I2C configuration, analog input, sensor data, relay, watchdog, and persistent configuration
Raw digital sample	0 to 4095	12-bit ADC code returned as an unsigned 16-bit little-endian value
ADC update interval	Approximately 20 ms	Background acquisition; an I2C read returns the latest published sample
Command processing delay	2 to 5 ms	Use 5 ms conservatively before reading
Pending setter timeout	250 ms	Parameter must arrive before expiry

These values describe the currently documented digital behavior. Detailed DDP commands and transaction sequences are provided in Chapter 3.

2.3 Onboard Interface Controller

The onboard controller acquires the analog output of the GT36537 LDR divider and provides the processed sensor data through the module's I2C interface. The controller is not exposed as a general-purpose microcontroller and is not intended for user programming.

Feature	Module implementation
Internal controller	32-bit Arm Cortex-M0+
Sensor acquisition	12-bit ADC
User digital interface	I2C
I2C addressing	7-bit addressing
Supported I2C clock	100 kHz and 400 kHz
Direct sensor access	Analog SIGNAL contact
Programming and debugging	Reserved for factory use

2.4 Module-Level Characteristics

The following characteristics are established for the current module implementation and are described in the preceding sections.

Characteristic	Current module specification
Nominal supply voltage	3.3 V or 5 V
Digital interface	I2C, 7-bit slave
Factory I2C address	0x26
Configurable I2C address range	0x08 to 0x77

Characteristic	Current module specification
I2C clock frequency	100 kHz to 400 kHz
Sensor acquisition	12-bit ADC
Raw digital output	0 to 4095
ADC update interval	Approximately 20 ms
Command processing delay	2 to 5 ms
Direct analog access	SIGNAL contact on the RAW Signal Header

Additional characteristics such as complete-module absolute maximum ratings, supply-current limits, guaranteed analog-output accuracy, calibrated lux accuracy, board-level operating-temperature range, and environmental qualification are not specified unless explicitly stated elsewhere in this Product Reference.

Component-level ratings shall not be interpreted as guaranteed ratings of the complete module.

2.5 Electrical Precautions

1. Use a current-limited 3.3 V or 5 V supply during engineering bring-up.
2. Verify connector orientation and establish a common ground before applying power.
3. Do not exceed the controller upper operating-voltage limit of 5.5 V.
4. When operating the module at 5 V, verify host compatibility with the actual voltage present on SDA, SCL, and SIGNAL. Use appropriate level translation when required.
5. During normal operation, use the I2C connection points only for I2C communication. The same physical signals are shared with the controller’s factory programming and debugging functions; SWD is not provided as a separate user interface.
6. Do not attempt firmware replacement or external debugging through the I2C connection points. Programming, debugging, and reset functions are reserved for manufacturer use and factory diagnostics.
7. Remove power before cutting, soldering, or reworking the I2C disable bridge or optional connector positions.
8. Verify the current released schematic, component specifications, and interface documentation before production integration.

3 Functional Overview

The DevLab I2C GT36537 LDR Ambient Light Sensor combines a GT36537 photoresistor (LDR), wired as a voltage divider, with an onboard controller. The module provides two ways to observe light level: digital readings over I2C and direct access to the analog sensor signal at SIGNAL.

The GT36537 LDR itself is an analog device. The onboard controller samples its output, applies the selected averaging, and provides the latest raw reading through DevLab Device Protocol (DDP) v1.0.

3.1 Functional Block Diagram

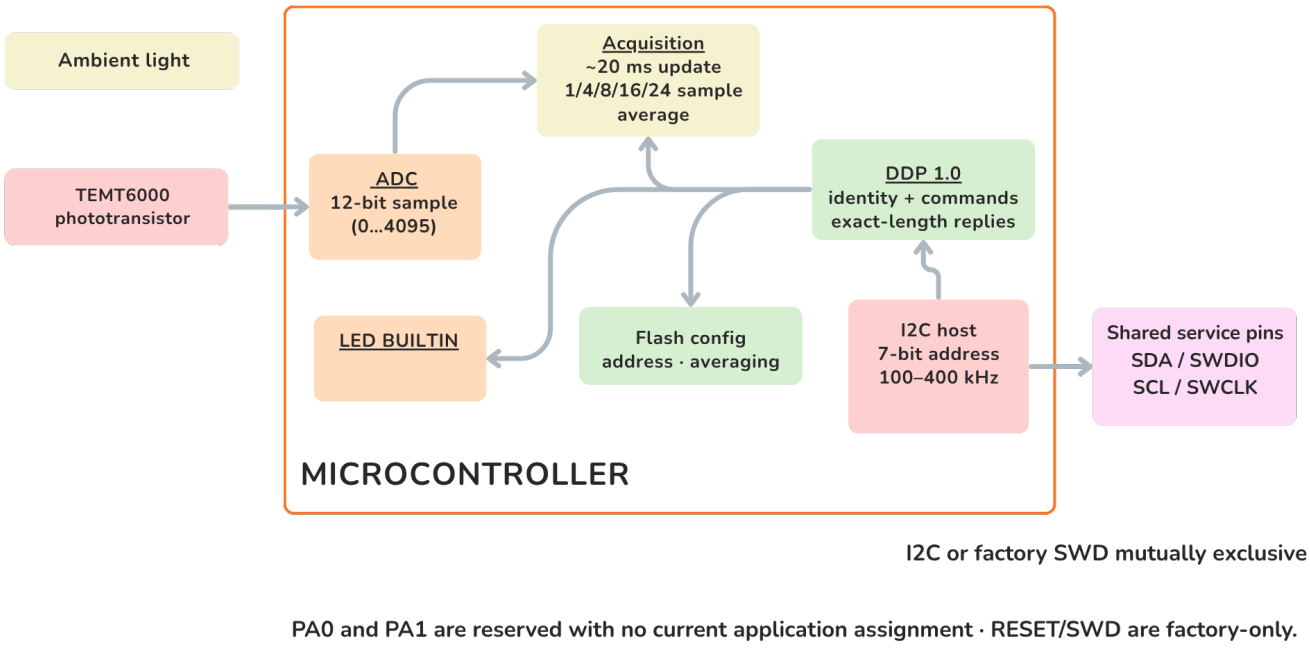


Figure 3.1 — Functional signal and control paths. Ambient light changes the LDR divider output. The signal is available at SIGNAL and is also acquired by the controller ADC for access over I2C.

3.2 Operating Modes

Mode	Connection	User-visible result
Digital I2C	VCC, GND, SDA, and SCL	Averaged 12-bit raw light reading through DDP
Direct analog	VCC, GND, and SIGNAL	Light-dependent voltage for a host ADC or measurement instrument

Both modes observe the same sensor path. An I2C read returns the latest available sample; it does not trigger a new light measurement. The direct analog output may be used independently for development, comparison, or application-specific calibration.

3.3 Digital Light Measurement

The onboard ADC produces an unsigned 12-bit value. Whether the value rises or falls with more light depends on which end of the voltage divider carries the LDR, under the actual supply, optical, and mechanical conditions.

Measurement property	Value
ADC resolution	12 bits
Raw output range	`0..4095`
Response format	Two bytes, `[LSB, MSB]`
Update interval	Approximately 20 ms
Selectable averaging	1, 4, 8, 16, or 24 samples

The averaging setting controls the balance between response speed and stability:

Samples	Approximate time for a completely new average	Typical use
1	20 ms	Fast changes
4	80 ms	Low-latency smoothing
8	160 ms	General monitoring
16	320 ms	Stable ambient-light monitoring
24	480 ms	Maximum available smoothing

The result is a relative ADC reading, not a calibrated lux measurement. Quantitative lux measurements require calibration in the final enclosure and optical arrangement.

3.4 I2C Communication Model

The I2C host always initiates communication. The 7-bit I2C address selects the module, while the one-byte DDP selector identifies the requested function. These are independent values.

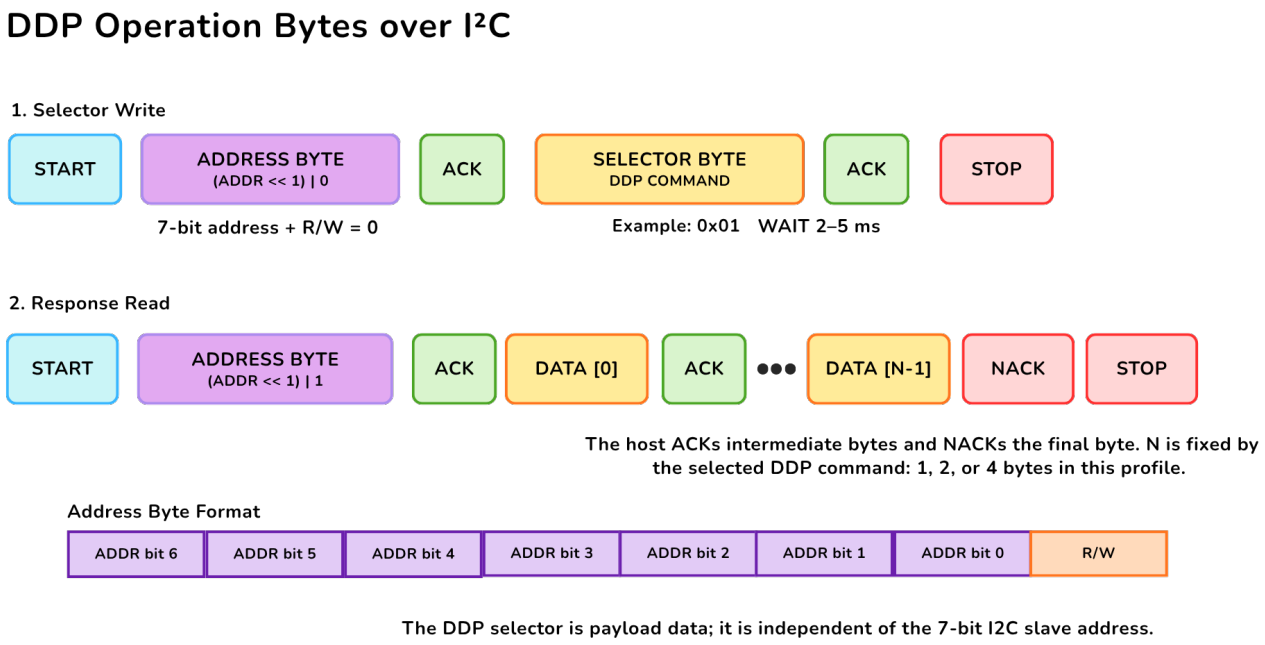


Figure 3.2 — DDP command and response over I2C. The host writes one DDP selector, issues STOP, waits 2–5 ms, and then reads the documented number of response bytes. The host acknowledges intermediate bytes and sends NACK after the final byte.

Communication property	Value
Factory 7-bit address	`0x26`
Configurable address range	`0x08..0x77`
I2C clock	100 kHz to 400 kHz
Delay before response read	2–5 ms; use 5 ms conservatively
Byte order for multi-byte values	Little endian
Supported response lengths	1, 2, or 4 bytes, depending on the selected function

Pass `0x26` directly to an I2C library. Values `0x4C` and `0x4D` are the shifted write/read address bytes visible on the bus, not the 7-bit address used by typical host APIs.

3.5 User-Accessible DDP Functions

User function	DDP selector	Result
Identify the module	`0x00`	Device ID `0x0106`

User function	DDP selector	Result
Read protocol version	`0x04`	DDP version 1.0
Read active I2C address	`0x20`	Current 7-bit address
Change I2C address	`0x21`	Stores and applies a new address in the valid range
Restore factory configuration	`0x23`	Restores default configuration; reset required
Read ambient-light data	`0x60`	Latest two-byte 12-bit raw sample
Set ADC averaging	`0x62`	Selects 1, 4, 8, 16, or 24 samples
Read ADC averaging	`0x63`	Returns the active averaging setting
Turn `BUILTIN` off/on	`0xA0` / `0xA1`	Drives the built-in indicator LOW/HIGH
Pulse `BUILTIN`	`0xA2`	Starts a non-blocking timed pulse
Set/read pulse time	`0xA3` / `0xA4`	Uses 1–40 units of 25 ms

The I2C address, averaging selection, and pulse duration are retained after a power cycle. Changing a stored value should be an occasional configuration operation rather than part of the normal measurement loop.

Detailed transaction handling is implemented by the DevLab_GT365xx Arduino library listed in Chapter 8, which uses DevLab Interface for I2C bus management.

Arduino Library Functions

All functions belong to the DevLab_GT365xx class. Functions that return `bool` return `true` on success. Readings are taken with the DDP read-ADC command (`0x60`) and are only available after a successful `begin` call.

Function	Description
`DevLab_GT365xx(wire, address, clock)`	Constructor. Defaults: `Wire`, address `0x26`, 400000 Hz
`begin()`	Starts the I2C bus on the host's default pins and verifies the GT36537 Device ID
`begin(sdaPin, sclPin, clock)`	Same as `begin()`, selecting the SDA/SCL pins and I2C clock
`beginRecovered(sdaPin, sclPin, timeoutUs, restart)`	Starts the bus with bus recovery before verifying the Device ID. Defaults: `timeoutUs` 20000, `restart` false
`readRaw(rawValue)`	Latest 12-bit ADC reading, `0` to `4095`
`readPercentage(percentage)`	Raw reading as a percentage, `raw × 100 / 4095`
`readInvertRaw(invertedValue)`	Inverted 12-bit reading, `4095 − raw`
`readInvertPercentage(percentage)`	Inverted reading as a percentage
`busReady()`	`true` if the I2C bus was started successfully
`isConnected()`	`true` if the GT36537 Device ID was verified at the configured address
`deviceInfo()`	Identity information read during `begin`
`printInfo(out)`	Prints the device information to a `Print` stream (default `Serial`)
`protocol()`	Access to the underlying DDP master for direct commands

Function	Description
bus ( )	Access to the underlying DevLab Interface bus object

The inverted functions are useful when the reading falls as light increases, which depends on which end of the voltage divider carries the LDR.

3.6 Built-In Indicator

The built-in status indicator is controlled through the DDP actuator functions. The shared DDP command names contain RELAY, but this product uses them for the built-in indicator and does not provide an electromechanical relay output.

The pulse duration is configurable from 25 ms to 1000 ms in 25 ms steps. A timed pulse runs without interrupting light measurements or I2C communication.

3.7 Physical Interface Notes

The SIGNAL contact provides direct access to the analog sensor path. Its complete board-level transfer function, source impedance, and guaranteed output range are not specified for the current revision. Use a high-impedance input and verify the actual range before connecting it to a lower-voltage ADC.

The PCB provides three I2C connection positions carrying GND, VCC, SDA, and SCL: one connector is populated by default and two horizontal positions are provided for optional 4-pin, 1.00 mm-pitch JST/Qwiic-compatible connectors. All three positions access the same I2C bus.

The I2C connection points share signals with the controller’s factory programming and debugging interface; there is no separate user-accessible SWD connector. Programming, debug, and reset functions are reserved for manufacturer programming and factory diagnostics, not user firmware replacement.

Two expansion pads are reserved and have no current user function. The I2C disable bridge is intended for applications that require the analog path without active I2C operation; remove power before cutting or restoring it.

4 Connectors & Pinouts

The user-accessible interfaces are described below by module function. The released documentation does not provide controlled contact numbers or a mating-connector part number, so the signal order must be checked against the connector orientation before producing a harness.

4.1 General Pinout

Connection group	Signals	Function
Direct sensor contacts	`VCC`, `GND`, `SIG`	Module power and direct access to the TEMT6000 analog signal
I2C connection positions	`GND`, `VCC`, `SDA`, `SCL`	Power and digital communication over the common I2C bus
Reserved expansion pads	Reserved	No current application assignment; electrical limits are not specified

The PCB has three physical I2C connection positions. One connector is populated by default, and two additional horizontal positions accept optional 4-pin, 1.00 mm-pitch JST/Qwiic-compatible connectors. All three positions are connected to the same I2C bus.

4.2 Signal and I2C Connection Guide

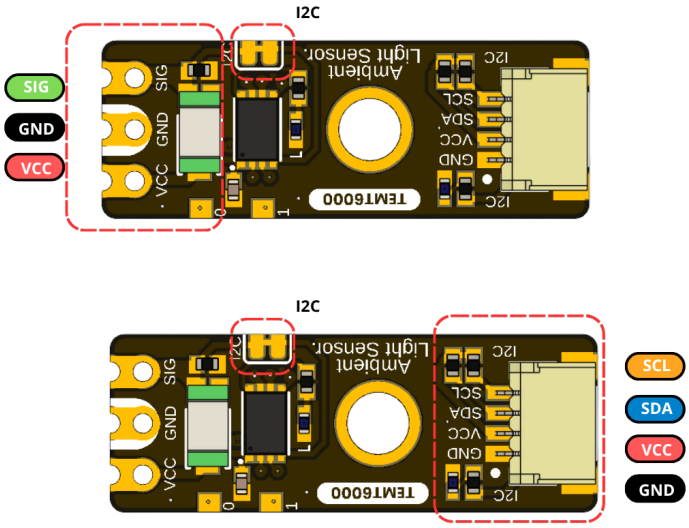


Figure 4.1 — Direct analog and I2C connection groups. The direct contacts provide SIG, GND, and VCC; the populated I2C connector provides SCL, SDA, VCC, and GND. Always follow the connector orientation shown on the board.

4.3 Direct Sensor Contacts

Label	Type	Description
VCC	Power	Module supply; nominal 3.3 V or 5 V operation
GND	Power	Common power and signal reference
SIG / SIGNAL	Analog	Direct light-dependent sensor signal; current transfer function unspecified

These three contacts are shown at the end opposite the populated I2C connector. SIG provides direct access to the TEMT6000 analog signal; it is not a separate specialized communication port.

4.4 I2C Connection Positions

Signal	Type	Description
GND	Power	Common return
VCC	Power	I2C peripheral supply; nominal 3.3 V or 5 V
SDA	Bidirectional I2C data	Host must tolerate the actual bus pull-up voltage

Signal	Type	Description
SCL	I2C clock	Supported at 100 kHz to 400 kHz

The two optional horizontal I2C connector positions use the same signal set and 1.00 mm pitch as the populated connector. Because all three positions are physical access points to the same bus, an optional connector may be used for an alternate cable orientation or bus continuation. Account for the loading and pull-ups of every attached device.

The I2C connection points share signals with the controller's factory programming and debugging interface. These functions are mutually exclusive, and there is no separate user-accessible SWD connector.

4.5 Auxiliary and Factory Service Functions

Function	Documented role	Qualification status
Reserved expansion pad 1	Reserved connection	No current application assignment; electrical limits unspecified
Reserved expansion pad 2	Reserved connection	No current application assignment; electrical limits unspecified
Controller reset	Factory service function	Manufacturer programming and factory diagnostics only

Programming, debugging, and reset are intended primarily for development, manufacturing, and service purposes. Currently, user firmware programming is not supported or documented. During programming or diagnostics, I2C must be inactive and other bus devices must be isolated from the shared signals.

4.6 Indicators and I2C Bridge

The module provides a power indicator, a firmware-controlled built-in status indicator, and a normally closed solder bridge that connects the sensor's raw output signal to the controller ADC input. Opening the bridge isolates the raw sensor signal from the ADC, and the connection can be restored by re-soldering the bridge. The controller ADC input used for sensor acquisition is internal to the module and is not an additional external contact.

5 Board Operation

This chapter shows how to verify one DevLab I2C GT36537 LDR Ambient Light Sensor and read its raw ambient-light value with Arduino. The examples use the DevLab_GT365xx library and support ESP32, RP2040/RP2350, and AVR hosts.

5.1 Required Hardware and Software

- DevLab I2C GT36537 module
- ESP32, RP2040/RP2350, or AVR development board acting as I2C host
- I2C connection for VCC, GND, SDA, and SCL
- USB cable for programming and Serial Monitor access
- Arduino IDE with the selected board package installed
- DevLab_GT365xx Arduino library

5.2 Library Installation

1. Open **Tools > Manage Libraries** in Arduino IDE.
2. Search for DevLab_GT365xx.
3. Select the current release and click **Install**.

The library implements DDP v1.0 and uses DevLab Interface for I2C bus management. Its dependencies, DevLabDDP and DevLab_Interface, are installed together with it, so the sketch only includes DevLab_GT365xx.h and does not need separate bus-handling or bus-recovery code.

5.3 I2C Connection and Example Settings

Connect the module with power removed. Align the connector with the orientation printed on the board and verify VCC, GND, SDA, and SCL before applying power. A connection diagram for this GT36537 revision is pending publication.

Module	ESP32	RP2040 / RP2350	AVR	Function
VCC	3.3 V or 5 V	3.3 V or 5 V	3.3 V or 5 V	Module supply
GND	GND	GND	GND	Common reference
SDA	GPIO6	GPIO12	Fixed board SDA pin	I2C data
SCL	GPIO7	GPIO13	Fixed board SCL pin	I2C clock

On RP2040 and RP2350 the examples use the `Wire1` bus. AVR boards have fixed I2C pins (Uno/Nano: A4/A5, Mega: 20/21, Leonardo: 2/3) taken from the board variant. The pin numbers above match the repository examples; change `SDA_PIN` and `SCL_PIN` when the selected host board uses different I2C pins.

Important: The I2C pull-up voltage follows the module supply (VCC). When operating the module at 5 V, verify that the host I2C pins are 5 V tolerant. If the host operates at 3.3 V and is not 5 V tolerant, use appropriate bidirectional level translation between the module and the host.

The example uses the following settings:

Constant	Default	Purpose
`SDA_PIN`	GPIO6 (ESP32); GPIO12 (RP2040/RP2350)	Host SDA pin
`SCL_PIN`	GPIO7 (ESP32); GPIO13 (RP2040/RP2350)	Host SCL pin
`I2C_FREQ`	400000 Hz	I2C clock
`SENSOR_ADDRESS`	`0x26`	Factory 7-bit module address
`READ_INTERVAL_MS`	250 ms	Time between printed readings

If the module address was changed previously, update `SENSOR_ADDRESS` before uploading the sketch.

5.4 Single-Sensor Arduino Example

Open `serialLightRead.ino` from the library examples, or create it with the following sketch:

```
/**
 * @file serialLightRead.ino
 * @brief Verifies a GT36537 DDP device (ID 0x0106) on the I2C bus and prints
 *        its raw ADC0 reading and relative percentage over Serial.
 */

#include <Arduino.h>
#include <Wire.h>
#include <DevLab_GT365xx.h>

#if defined(ARDUINO_ARCH_RP2040) || defined(ARDUINO_ARCH_RP2350)
  #define I2C_BUS Wire1
  constexpr int SDA_PIN = 12;
  constexpr int SCL_PIN = 13;
#elif defined(ARDUINO_ARCH_ESP32)
  #define I2C_BUS Wire
  constexpr int SDA_PIN = 6;
  constexpr int SCL_PIN = 7;
#elif defined(ARDUINO_ARCH_AVR)
  #define I2C_BUS Wire
  constexpr int SDA_PIN = SDA;
  constexpr int SCL_PIN = SCL;
#else
#error "Use an ESP32, RP2040, RP2350, or AVR master"
#endif

constexpr uint32_t I2C_FREQ = 400000UL;
constexpr uint8_t SENSOR_ADDRESS = 0x26U;
constexpr uint32_t READ_INTERVAL_MS = 250U;

DevLab_GT365xx sensor(I2C_BUS, SENSOR_ADDRESS, I2C_FREQ);

void setup() {
  Serial.begin(9600);

  sensor.beginRecovered(SDA_PIN, SCL_PIN);
  if (!sensor.busReady()) {
    Serial.println("ERROR: I2C bus is blocked");
    return;
  }

  if (!sensor.isConnected()) {
    Serial.println("ERROR: expected GT36537 DDP device ID 0x0106 at 0x26");
    return;
  }

  sensor.printInfo();
  Serial.println("GT36537 ready: PA2 ADC0 reads the LDR divider");
}

void loop() {
  static uint32_t lastRead = 0U;
  if (millis() - lastRead < READ_INTERVAL_MS) {
    return;
  }
  lastRead = millis();

  uint16_t raw = 0U;
  if (sensor.readRaw(raw)) {
    float percent = (raw * 100.0f) / 4095.0f;
    Serial.printf("adc0=%u percent=%.1f\n", raw, percent);
  } else {
    Serial.println("ERROR: light read failed");
  }
}
```

The complete example also supports AVR hosts, where `Serial.printf()` is not available and the output is printed with `Serial.print()` instead.

The following companion examples are provided:

Example	Purpose
`serialLightReadInverted`	Reports `4095 - raw` with `readInvertRaw()` and `readInvertPercentage()` so that the value rises with more light
`serialLightRawPlot`	Streams one raw integer per line at 115200 baud for the Arduino Serial Plotter or `python/serial_plotter.py`
`i2c_scanner`	Scans the I2C bus at 115200 baud and prints the 7-bit address of every device that responds; a detected address only confirms that a device acknowledged, not that it is a GT36537
`changeAddress`	Serial tool at 115200 baud to scan for DDP devices and reassign the module address (factory address `0x26`), for example `scan` or `change 26 30`

5.5 Upload and Serial Output

1. Select the connected host board in Arduino IDE.
2. Select its serial port.
3. Compile and upload `serialLightRead.ino`.
4. Open Serial Monitor at **9600 baud** (115200 baud for `serialLightRawPlot`).
5. Illuminate and cover the LDR to confirm that the printed value changes.

At startup, the sketch recovers and initializes the I2C bus, verifies that address `0x26` contains a DDP GT36537 with Device ID `0x0106`, and prints the reported device information. It then prints `adc0` and `percent` every 250 ms.

```
<DDP device information>
GT36537 ready: PA2 ADC0 reads the LDR divider
adc0=<value from 0 to 4095> percent=<0.0 to 100.0>
adc0=<value from 0 to 4095> percent=<0.0 to 100.0>
```

The readings are raw ADC values. They indicate relative light level and must not be interpreted directly as lux. Whether the value rises or falls with more light depends on which end of the voltage divider carries the LDR; use `serialLightReadInverted` if a value that rises with light is preferred.

5.6 Troubleshooting

Serial output or symptom	Check
`ERROR: I2C bus is blocked`	Wiring, connector orientation, shared ground, pull-up voltage, and unintended SDA/SCL drive
`ERROR: expected GT36537 DDP device ID 0x0106 at 0x26`	Address setting, I2C pins, module power, and possible address collision
`ERROR: light read failed` after successful identification	Cable continuity, bus clock, supply stability, and I2C connections
Constant `0` or `4095`	Sensor obstruction, excessive illumination, supply, and module orientation
Value falls when light increases	Expected for some divider orientations; use `serialLightReadInverted`
Values change but lux is inaccurate	Calibrate the complete optical installation against a reference light meter

The complete Arduino examples and the DevLab_GT365xx library are listed in Chapter 8.

6 Mechanical Information

The current mechanical drawing defines the board outline and the dimensions shown below. Dimensions are in millimetres.

6.1 Board Dimensions

Figure 6.1 — Board dimensions.

Use the drawing itself as the dimensional reference; do not derive additional dimensions by scaling the image.

6.2 Integration Clearances

Leave optical clearance around the GT36537 LDR sensing surface and its field of view. Provide connector insertion and cable bend clearance at the populated I2C connector and at any optional connector position that will be assembled. Keep conductive fasteners away from contacts and factory service points, and avoid enclosure features that shadow the sensor or indicators.

6.3 Mechanical Data Not Specified

- Board thickness, dimensional tolerances, and finished mass
- Unannotated feature coordinates and edge clearances
- Optional connector population variants and mating-part numbers
- Connector insertion envelope, retention force, and cable bend radius
- Maximum top- and bottom-side component heights
- Optical axis, keep-out cone, and enclosure-window specification
- Reserved expansion-pad and factory service-point coordinates

7 Company Information

Item	Information
Company / brand	UNIT Electronics
Board ecosystem	DevLab
Product family	Atom
Website	UNIT Electronics
Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX, Mexico

The product repository and controlled technical references are listed in the next chapter.

8 Reference Documentation

Reference	Location
Board topology	–
Board dimensions	–
Product repository	UNIT-Electronics-MX/unit_devlab_i2c_ldr_photoresistor_light_sensor
Hardware reference	hardware/README.md
English pinout	–
Spanish pinout	–
Board top view	–
Board bottom view	–
Current schematic	–
Software examples	C++ (Arduino)
Arduino library	UNIT-Electronics-MX/unit_devlab_gt365xx_library
Local DDP device profile	GT36537 DDP profile
Canonical DDP definitions and Arduino library	UNIT-Electronics-Labs/unit_devlab_ddp_library
Component page	–
GT36537 component datasheet	–

The released pinout, current board images, topology, dimensions, and schematic define the current visible interfaces, board layout, and mechanical envelope once they are published for this GT36537 revision. The component datasheet does not define complete-module ratings. The DDP profile defines runtime identity and digital data transport; it does not replace module-level electrical qualification.

9 Appendix

9.1 Current Schematic

The schematic documents the controller-based board, including the reference controller, LDR analog stage, I2C interfaces, status indicators, test points, and optional connectors.

Figure 9.1 — Electrical schematic.

The LDR divider signal is routed to the controller ADC input and the external VCC/GND/SIGNAL header. The reading is not a calibrated or guaranteed conversion to lux.

The I2C SDA and SCL signals are available at three physical connection positions: one connector populated by default and two optional horizontal positions. These signals are shared with the controller's factory programming and debugging interface; there is no separate user-accessible SWD connector.

9.2 Document Control

Field	Value
Product	DevLab I2C GT36537 LDR Ambient Light Sensor
Product hierarchy	UNIT Electronics → DevLab board ecosystem → Atom family
Manufacturer Part Number (MPN)	UE0164
Current board artwork	–
Current pinout, topology, and dimensions	–
Available schematic	–
Product Reference	Version 1.0.0
Publication date	2026-08-31
Interfaces	Three I2C connection positions on one bus; VCC/GND/SIGNAL header; reserved expansion pads; shared factory programming/ debug signals

9.3 Technical Data Not Specified in This Revision

- Complete-module absolute-maximum ratings and current consumption
- I2C pull-up values and validated bus-loading limits
- Direct SIGNAL output range, loading characteristics, and ADC guidance
- Mechanical tolerances, board thickness, component heights, and finished mass
- Module-level optical, electrical, environmental, and EMC characterization